Dhritabrata Mitra
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EDUCATION

M.S. in Computer Science, Michigan Technological University, GPA: 3.85/4.0 — Aug 2023 – Apr 2025

B.Tech in Computer Science, University of Engineering & Management, GPA: 3.39/4.0 — Jul 2016 – May 2020

WORK EXPERIENCE

Cognizant, *Software Engineer II* — Jan 2020 – Jun 2023

Client: Netflix

Software Engineer II — Jan 2021 – Jun 2023

- Built an **Entitlement & Regional Catalog microservices** using Spring Boot **REST APIs** on AWS to determine content availability based on user region and licensing policies; supported 90M+ users with P95 ≤150ms latency and 99.95% uptime, growing a $30B+ streaming platform.
- Saved **$200,000/year** by enhancing content search functionality by tuning Elasticsearch queries, adding Redis caching, optimizing PostgreSQL schema and indexes, and JPA queries, reducing search latency by 40%.
- Modified an **event-driven Kafka pipeline** across distributed systems to address delayed and duplicate notifications by implementing idempotent producers, exponential backoff retries, and consumer deduplication, reducing latency by 45% and duplicate incidents by 95%.
- Containerized React/Next.js frontend and Java services with **Docker** and orchestrated deployments on **AWS EKS** (EC2-backed) using Helm charts, allowing blue/green, rolling, and canary releases, cutting release cycles by 50%.
- Reduced manual deployment errors by 99% by **implementing Jenkins CI/CD pipelines** to stabilize production deployments, integrating with AWS CodeDeploy, configuring IAM, and adding AWS Lambda notifications; automated build, test, and release validation.
- Implemented **IaC governance** for **Kubernetes** and **Terraform** (policy checks and gated CI), blocking insecure configs before deploy and reducing security/ops exceptions by 40% while improving deployment success to 99%.

Software Engineer I — Jan 2020 – Dec 2020

- Resolved frontend and backend production issues through root cause analysis, reducing downtime from **12 hours per quarter to 30 minutes** and cutting average production incidents from 250 to 20 per month.
- Eliminated over-fetching, inconsistent pagination, and fragmented responses by optimizing **REST** and **GraphQL API** integrations and standardizing API responses, reducing page errors by 30% and improving responsiveness by 60%.
- Built **CloudWatch** and **Splunk dashboards** and alerts for Java and Kafka services, and drove runbooks and RCA follow-ups, cutting MTTD by 40%, MTTR by 25%, and recurring defects by 20% during Sev-2/Sev-3 incidents.
- Built and maintained unit, integration, and end-to-end test suites using **JUnit**, **Mockito**, **Jest**, and **Playwright**, following Test-Driven Development (TDD) practices, improving backend reliability by 50% and production issues by 70%.

CURRENT RESEARCH EXPERIENCE

Michigan Technological University, *Research Assistant* — Jun 2025 – Present

- Built and iterated **AI/ML prototypes** using **Python**, translating research ideas into working systems and validating real-world applicability across multiple experimental runs and datasets.
- Designed **evaluation frameworks** and **KPI-driven benchmarks** to measure model performance (accuracy, latency, reliability), improving experiment comparability and decision-making speed by 35%.
- Developed **automated data pipelines** for experimentation and analysis, reducing manual preprocessing and evaluation effort by 40% and accelerating iteration cycles.
- Collaborated with researchers and stakeholders to define **problem statements, analyze large-scale datasets,** and **deliver insights** that influenced research direction across multiple projects.

SKILLS

Languages: Java, Python, TypeScript, SQL, JavaScript
Backend: Spring Boot, Spring MVC, Spring Data JPA, Spring Security, REST APIs, GraphQL APIs, Node.js
SQL/NoSQL Databases: PostgreSQL, Cassandra, Redis, Elasticsearch, MongoDB
Event Streaming: Apache Kafka, RabbitMQ
Cloud/DevOps: AWS (EC2/EKS/S3/Lambda/IAM), Docker, Kubernetes, Terraform, Helm, Jenkins, Maven, Git, Linux
Observability: AWS CloudWatch, Apache Splunk
Testing: JUnit, Mockito, Jest, Playwright
Frontend: React.js, Redux, Zustand, Tailwind CSS, HTML5, CSS3
Realtime Communication: WebRTC, WebSockets, Socket.IO

TECHNICAL PROJECTS

Meet AI | Demo | Live Site | GitHub

- Built a live AI agent that joins video calls using OpenAI's Realtime API, delivering sub-second voice responses with a natural conversational feel.
- Eliminated manual mid-meeting searches by enabling direct agent Q&A, reducing answer retrieval from 3–5 minutes to under 3 seconds.
- Designed agent-based orchestration with Inngest Agent Kit to automate multi-step post-meeting workflows.
- Built a pipeline combining transcript fetching, parsing, speaker enrichment, and LLM-based summarization.
- Engineered dynamic system prompt generation, adapting per meeting context, agent instructions, and conversation state.
- Built with Next.js, Nginx, OAuth, JWT, Tailwind CSS, WebRTC, tRPC, Drizzle ORM, and Polar API.

Plant Disease Detection | Demo | GitHub

- Built a full-stack AI application (React.js and Python/FastAPI) to classify leaf images as Early Blight, Late Blight, or Healthy, enabling sub-2s end-to-end predictions for non-technical users.
- Designed and trained Python-based CNN models in TensorFlow/Keras, experimenting with 3+ architectures (custom CNN, and transfer learning), and achieved the best test accuracy of 95%
- Built an end-to-end Python ML pipeline (data to model to REST API to web UI), cutting model iteration time by 50% through reusable preprocessing and repeatable training runs.
- Built a Python FastAPI inference endpoint delivering single-image predictions in 150 ms (p95) with top-1 label and confidence, deploying a lightweight exported Keras (.h5) model optimized for low-memory and edge inference.